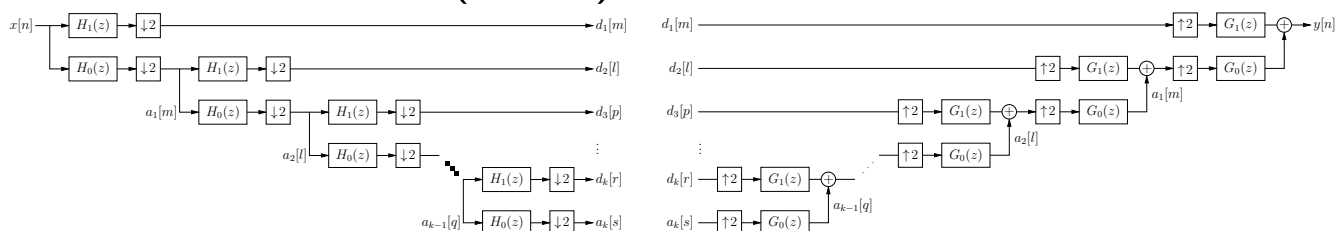
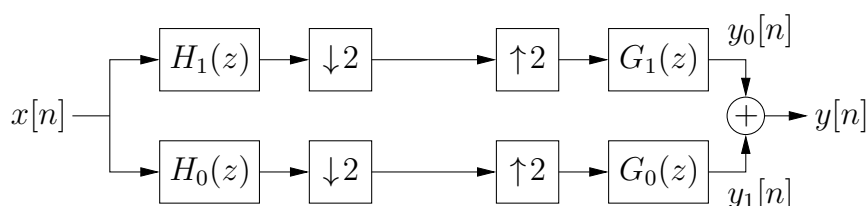


2-Channel PR-FIR Filterbanks:

- In the last note packet, we introduced the discrete wavelet transform (DWT):



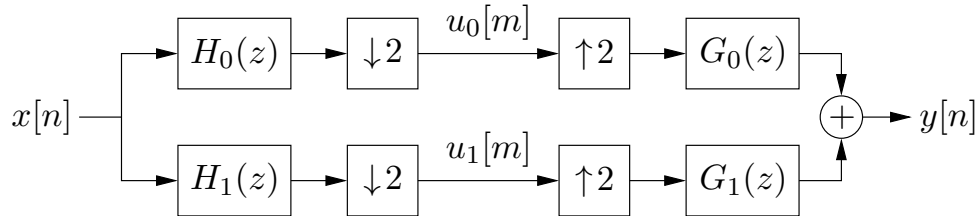
which is based on repeated application of a 2-channel filterbank with the form



- If that filterbank gives perfect reconstruction (i.e., $y[n] = x[n - \ell]$ for some $\ell \in \mathbb{Z}$), then so does the DWT.
- And if that filterbank splits $x[n]$ into two orthogonal subspaces (i.e., $y_0[n]$ is orthogonal to $y_1[n]$ for any $x[n]$), then the DWT will be an orthogonal transform.
- In this note packet, we will learn how to design perfect-reconstruction FIR filterbanks, both orthogonal and biorthogonal.
- Biorthogonal is useful because (as we'll see) orthogonal PR-FIR designs can't be linear phase.

Aliasing Cancellation:

- Consider the following 2-channel filterbank:



where usually $H_0(z)$ is lowpass and $H_1(z)$ is highpass.

- The intermediate signals can be written as

$$U_i(z) = \frac{1}{2} \sum_{p=0}^1 H_i(z^{\frac{1}{2}} e^{-j\pi p}) X(z^{\frac{1}{2}} e^{-j\pi p}) \quad \text{for } i = 0, 1$$

$$\begin{aligned} Y(z) &= \sum_{i=0}^1 G_i(z) U_i(z^2) \\ &= \sum_{i=0}^1 G_i(z) \frac{1}{2} \sum_{p=0}^1 H_i(z e^{-j\pi p}) X(z e^{-j\pi p}) \\ &= \frac{1}{2} \sum_{i=0}^1 G_i(z) (H_i(z) X(z) + H_i(-z) X(-z)) \\ &= \frac{1}{2} \begin{bmatrix} X(z) & X(-z) \end{bmatrix} \begin{bmatrix} H_0(z) & H_1(z) \\ H_0(-z) & H_1(-z) \end{bmatrix} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix}. \end{aligned}$$

- Note that $X(-z)$ corresponds to $X(e^{j(\omega-\pi)})$ in the DTFT domain, i.e., an *aliased* version of $X(e^{j\omega})$.
- We don't want $X(-z)$ present in the output $Y(z)$!

Perfect Reconstruction FIR:

- For perfect reconstruction, we need $y[n] = x[n - \ell]$ for some integer ℓ . This, together with aliasing cancellation, implies

$$\begin{bmatrix} 2z^{-\ell} \\ 0 \end{bmatrix} = \underbrace{\begin{bmatrix} H_0(z) & H_1(z) \\ H_0(-z) & H_1(-z) \end{bmatrix}}_{\triangleq \mathbf{H}(z)} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix},$$

where $\mathbf{H}(z)$ is known as the “aliasing component matrix.”

- Equivalently,

$$\begin{aligned} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} &= \mathbf{H}^{-1}(z) \begin{bmatrix} 2z^{-\ell} \\ 0 \end{bmatrix} \\ &= \frac{1}{\det(\mathbf{H}(z))} \begin{bmatrix} H_1(-z) & -H_1(z) \\ -H_0(-z) & H_0(z) \end{bmatrix} \begin{bmatrix} 2z^{-\ell} \\ 0 \end{bmatrix} \\ &= \frac{2}{\det(\mathbf{H}(z))} \begin{bmatrix} z^{-\ell} H_1(-z) \\ -z^{-\ell} H_0(-z) \end{bmatrix} \end{aligned}$$

where

$$\det(\mathbf{H}(z)) = H_0(z)H_1(-z) - H_0(-z)H_1(z).$$

- We will assume that $H_0(z)$ and $H_1(z)$ are both real-coefficient FIR.
- Then, for $G_0(z)$ and $G_1(z)$ to be real-coefficient FIR, we need

$$\det(\mathbf{H}(z)) = cz^{-k} \quad \text{for } c \in \mathbb{R} \text{ and } k \in \mathbb{Z}.$$

Perfect Reconstruction FIR (cont.):

- Under the above determinant condition, we have

$$\begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} = \frac{2z^{-(\ell-k)}}{c} \begin{bmatrix} H_1(-z) \\ -H_0(-z) \end{bmatrix}$$

- Assuming $H_0(z)$ and $H_1(z)$ are causal with non-zero initial coefficient, we choose $k = \ell$ to keep $G_0(z)$ and $G_1(z)$ causal and free of unnecessary delay.
- And, we'll use $c = 2$ so that the gains of $G_i(z)$ and $H_i(z)$ are balanced.
- In summary, the 2-channel FIR perfect-reconstruction (PR) conditions are

$$\begin{aligned} H_0(z) \text{ \& } H_1(z) &: \text{ causal real-coefficient FIR} \\ \det(\mathbf{H}(z)) &= 2z^{-\ell} \text{ for } \ell \in \mathbb{Z}^+ \\ G_0(z) &= H_1(-z) \\ G_1(z) &= -H_0(-z). \end{aligned}$$

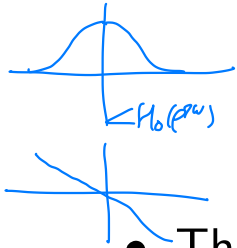
- Next, we will design the analysis filters, $H_0(z)$ and $H_1(z)$.
- Given the analysis filters, we can easily compute the synthesis filters, $G_0(z)$ and $G_1(z)$.

Orthogonal PR-FIR — Construction:

- The “orthogonal” PR-FIR design uses **causal**, **real-coefficient**, **even-length- L** filters $H_0(z)$ and $H_1(z)$ such that

$$2 = H_0(z)H_0(z^{-1}) + H_0(-z)H_0(-z^{-1}) \quad (1a)$$

$$H_1(z) = -z^{-(L-1)}H_0(-z^{-1}). \quad (1b)$$



- This satisfies the PR-FIR requirements because

$$\det(\mathbf{H}(z)) = H_0(z)H_1(-z) - H_0(-z)H_1(z)$$

$$\mathbf{H}(z) = \begin{bmatrix} H_0(z) & H_1(z) \\ H_0(-z) & H_1(-z) \end{bmatrix}$$

$$= z^{-(L-1)} (H_0(z)H_0(z^{-1}) + H_0(-z)H_0(-z^{-1}))$$

$$= 2z^{-(L-1)}, \text{ where we see } \ell = L-1.$$

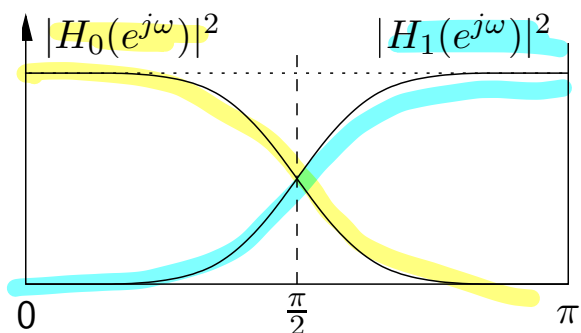
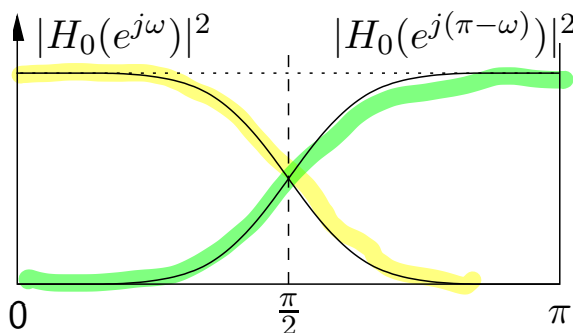
$h_0[n]$ real-valued
 $\Rightarrow |H_0(e^{j\omega})|$ even sym
 $\angle H_0(e^{j\omega})$ odd sym

- Plugging $z = e^{j\omega}$ into (1a), we see that

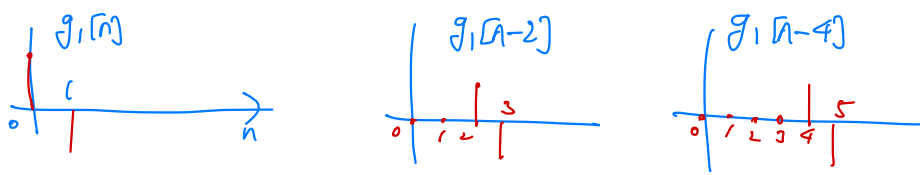
$$2 = H_0(e^{j\omega})H_0(e^{-j\omega}) + H_0(e^{j(\omega-\pi)})H_0(e^{-j(\omega-\pi)})$$

$$= |H_0(e^{j\omega})|^2 + |H_0(e^{j(\pi-\omega)})|^2 \text{ since } H_0(e^{-j\omega}) = H_0^*(e^{j\omega}),$$

which means that $H_0(e^{j\omega})$ is “power symmetric”:



- Also, (1b) implies $|H_1(e^{j\omega})| = |H_0(e^{j(\pi-\omega)})|$, which means that $H_0(e^{j\omega})$ and $H_1(e^{j\omega})$ are “power complementary.”



Orthogonal PR-FIR — Orthogonality:

- Under (1), the filters $h_0[n]$, $h_1[n]$, $g_0[n]$, and $g_1[n]$ are each orthogonal to their two-shifts. For example,

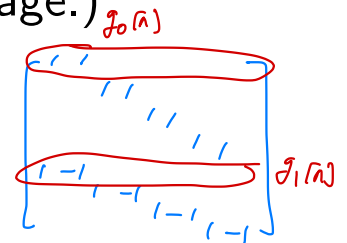
$$g_1[n] \cdot g_1[n-2m] = \sum_{n \in \mathbb{Z}} \underbrace{g_1[n]}_{h_1} \underbrace{g_1[n-2m]}_{h_1} = \delta[m], \quad m=0, \quad m \neq 0 \quad (2)$$

and similar for the other filters. (Proof on next page.)

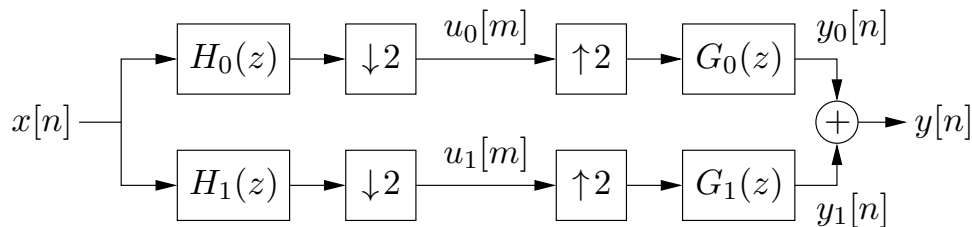
- In addition, it can be shown that (1) implies $W = \frac{1}{\sqrt{2}}$

$$W^T W = I$$

$$\sum_{n \in \mathbb{Z}} \underbrace{g_0[n]}_{h_0} \underbrace{g_1[n-2m]}_{h_1} = 0 \quad \forall m,$$



which then implies that the synthesis-filter output sequences $y_0[n]$ and $y_1[n]$ are themselves orthogonal.



- To see why $y_0[n]$ and $y_1[n]$ are orthogonal, first notice that

$$y_i[n] = \sum_{m \in \mathbb{Z}} u_i[m] g_i[n - 2m] \quad \text{for } i = 0, 1.$$

Then

$$\begin{aligned} \sum_{n \in \mathbb{Z}} y_0[n] y_1[n] &= \sum_{n \in \mathbb{Z}} \left(\sum_{m \in \mathbb{Z}} u_0[m] g_0[n - 2m] \right) \left(\sum_{k \in \mathbb{Z}} u_1[k] g_1[n - 2k] \right) \\ &= \sum_{m \in \mathbb{Z}} u_0[m] \sum_{k \in \mathbb{Z}} u_1[k] \underbrace{\sum_{n \in \mathbb{Z}} g_0[n - 2m] g_1[n - 2k]}_{= 0 \quad \forall m, k} = 0. \end{aligned}$$

Orthogonal PR-FIR — Proof: $a_1[k] = g_1[n] * g_1[n-k] \Big|_{n=k}$

- To see why the design (1) implies (2), we first write (2) as

$$a_1[2m] = \delta[m] \quad \text{with} \quad a_1[k] \triangleq \sum_{n \in \mathbb{Z}} g_1[n] g_1[n-k].$$

- In the z -domain, $a_1[k]$ can be written as

$$\begin{aligned} A_1(z) &= \sum_{k \in \mathbb{Z}} \left(\sum_{n \in \mathbb{Z}} g_1[n] g_1[n-k] \right) z^{-k} \\ &= \sum_{n \in \mathbb{Z}} g_1[n] z^{-n} \sum_{k \in \mathbb{Z}} g_1[n-k] z^{(n-k)} \\ &= \sum_{n \in \mathbb{Z}} g_1[n] z^{-n} \sum_{p \in \mathbb{Z}} g_1[p] (z^{-1})^{-p} \quad \text{via } p \triangleq n-k \\ &= G_1(z) G_1(z^{-1}). \end{aligned}$$

auto correlation of $g_1[n]$

- Likewise, the condition “ $a_1[2m] = \delta[m]$ ” can be written as

$$\frac{1}{2} \sum_{p=0}^1 A_1(z^{\frac{1}{2}} e^{-j\pi p}) = 1.$$

Since this must hold for all $z \in \mathbb{C}$, we can also write it as

$$\begin{aligned} 1 &= \frac{1}{2} \sum_{p=0}^1 A_1(z e^{-j\pi p}) = \frac{1}{2} [A_1(z) + A_1(-z)] \\ &= \frac{1}{2} [G_1(z) G_1(z^{-1}) + G_1(-z) G_1(-z^{-1})]. \end{aligned}$$

- Plugging in the PR-FIR condition $G_1(z) = -H_0(-z)$, we get

$$2 = H_0(-z) H_0(-z^{-1}) + H_0(z) H_0(z^{-1}),$$

which is identical to (1a).

Daubechies Orthogonal PR-FIR Design:

- If we design any one of the filters $h_0[n]$, $h_1[n]$, $g_0[n]$, $g_1[n]$, we can easily get the other three. We'll design the LPF $g_0[n]$.
- The usual approach is to design the “autocorrelation”

$$a_0[k] = \sum_{n \in \mathbb{Z}} g_0[n]g_0[n-k] \quad \text{s.t.} \quad a_0[2m] = \delta[m],$$

(recall p. 7), which can be restated in the z -domain as

$$A_0(z) = G_0(z)G_0(z^{-1}) \quad \text{s.t.} \quad A_0(z) + A_0(-z) = 2,$$

and then use “spectral factorization” to extract $G_0(z)$.

- There are many valid $\{a_0[k]\}$. We'll describe Daubechies' design, since the resulting filters are the most common choice for the orthogonal DWT.

- Daubechies chooses $G_0(z) = (1 + z^{-1})^{\frac{L}{2}} R(z)$, with causal order- $\frac{L-2}{2}$ $R(z)$ designed to satisfy the $A_0(z)$ condition. This puts the max # of zeros at $z = -1$ (i.e., $e^{j\omega} = e^{j\pi}$ or $\omega = \pi$).

- With this $G_0(z)$, we have $\stackrel{\Delta}{=} Q(z)$

$$A_0(z) = (1 + z^{-1})^{\frac{L}{2}} (1 + z)^{\frac{L}{2}} \overbrace{R(z)R(z^{-1})},$$

with i) symmetric $q[k]$, since $Q(z) = Q(z^{-1})$, and

ii) $Q(e^{j\omega}) \geq 0$, since $Q(e^{j\omega}) = R(e^{j\omega})R(e^{-j\omega}) = |R(e^{j\omega})|^2$.

- We'll first design $Q(z)$ and then use spectral factorization to extract $R(z)$ from $Q(z)$.

Daubechies Orthogonal PR-FIR (cont.):

- It helps to do a concrete example. Let's tackle $L = 4$.
- Since $R(z)$ is real-valued, causal, and order $\frac{L-2}{2} = 1$, $Q(z) = R(z)R(z^{-1})$ has the form

$$Q(z) = q_1 z + q_0 + q_1 z^{-1} \text{ with } q_0, q_1 \in \mathbb{R}.$$

- In this case,

$$\begin{aligned} A_0(z) &= (1 + z^{-1})^2 (1 + z)^2 \overbrace{(q_1 z + q_0 + q_1 z^{-1})}^{Q(z)} \\ &= (z^{-2} + 4z^{-1} + 6 + 4z + z^2)(q_1 z + q_0 + q_1 z^{-1}) \\ &= q_1(z^3 + z^{-3}) + (4q_1 + q_0)(z^2 + z^{-2}) \\ &\quad + (7q_1 + 4q_0)(z + z^{-1}) + (8q_1 + 6q_0) \\ &= \sum_{k \in \mathbb{Z}} a_0[k] z^{-k}. \end{aligned}$$

$$\begin{aligned} \delta[n] &\leftrightarrow 1 \\ \delta[n-1] &\leftrightarrow z^{-1} \\ \delta[n+1] &\leftrightarrow z \\ \delta[n-2] &\leftrightarrow z^{-2} \\ &\vdots \end{aligned}$$

$$\begin{aligned} a_0[0] &= 8q_1 + 6q_0 \\ a_0[1] &= a_0[-1] = 4q_1 + q_0 \\ a_0[2] &= a_0[-2] = 7q_1 + 4q_0 \\ a_0[3] &= a_0[-3] = q_1 \end{aligned}$$

- Since $a_0[2m] = \delta[m]$, we have

$$a_0[0] = 1 = 8q_1 + 6q_0$$

$$a_0[2] = 0 = 4q_1 + q_0.$$

Solving these equations yields $q_1 = -\frac{1}{16}$ and $q_0 = \frac{1}{4}$, so that

$$Q(z) = \frac{1}{4} \left(-\frac{1}{4} z^{-1} + 1 - \frac{1}{4} z \right).$$

- Finally, let's check the property " $Q(e^{j\omega}) \geq 0$." In our case,

$$Q(e^{j\omega}) = \frac{1}{4} \left(-\frac{1}{4} e^{-j\omega} + 1 - \frac{1}{4} e^{j\omega} \right) = \frac{1}{4} \left(1 - \frac{1}{2} \cos(\omega) \right) > 0,$$
 and so we have a valid $Q(z)$.

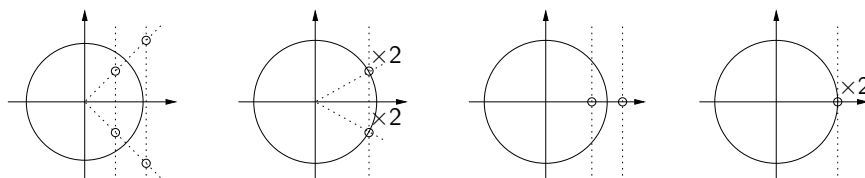
Daubechies Orthogonal PR-FIR (cont.):

- The last step is to extract real-coefficient, causal $R(z)$ from $Q(z) = R(z)R(z^{-1})$, known as “spectral factorization”.
- The factorization of $Q(z)$ takes the form

$$Q(z) = \underbrace{r_0 \prod_{n=1}^{\frac{L-2}{2}} (1 - \alpha_n z^{-1})}_{R(z)} \times \underbrace{r_0 \prod_{n=1}^{\frac{L-2}{2}} (1 - \alpha_n z)}_{R(z^{-1})},$$

with $\{\alpha_n\}$ the roots of $R(z)$ and $\{\alpha_n^{-1}\}$ the roots of $R(z^{-1})$.

- Because $R(z)$ has real-valued coefficients, α_n are either real or come in complex-conjugate pairs. (Same with α_n^{-1} .)
- Note that $\{\alpha_n^{-1}\}$ are reflections of $\{\alpha_n\}$ across the unit circle.
- In total, $Q(z)$ has $L - 2$ roots that have symmetry both across the real line and across the unit circle.



- To construct $R(z)$, we factor $Q(z)$. Then we keep all roots *inside* the unit circle, and one of each pair *on* the unit circle. This $R(z)$ is the “minimum phase” spectral factor of $Q(z)$.
- Finally, we solve for r_0 using the facts $Q(1) = \sum_k q[k]$ and $Q(1) = r_0^2 \prod_{n=1}^{\frac{L-2}{2}} (1 - \alpha_n)^2$, since we know $\{q[k]\}$ and $\{\alpha_n\}$.

Spectral factorization

$$H(e^{j\omega}) = \underbrace{|H(e^{j\omega})|}_{\text{magnitude}} \underbrace{e^{j\angle H(e^{j\omega})}}_{\text{phase}}$$

$$|H(e^{j\omega})| \longrightarrow H(e^{j\omega}) ?$$

"minimum phase"

stable & causal

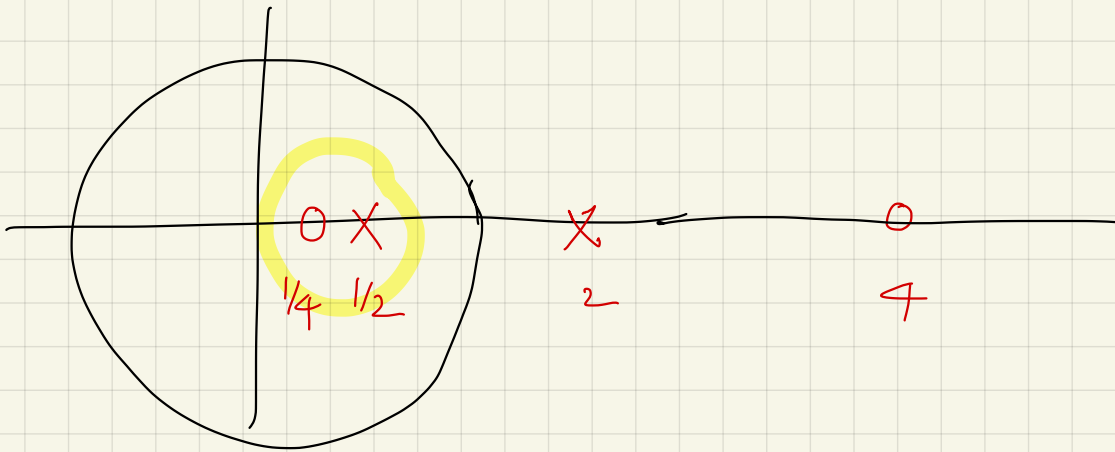
$\Rightarrow$ stable & causal inverse

example) $|H(e^{j\omega})|^2 = \frac{\frac{17}{16} - \frac{1}{2}\cos\omega}{\frac{5}{4} - \cos\omega}$

$$= \frac{\frac{17}{16} - \frac{1}{4}e^{j\omega} - \frac{1}{4}e^{-j\omega}}{\frac{5}{4} - \frac{1}{2}e^{j\omega} - \frac{1}{2}e^{-j\omega}} = H(e^{j\omega}) H(e^{-j\omega})$$

$$G(z) = H(z) H(1/z)$$

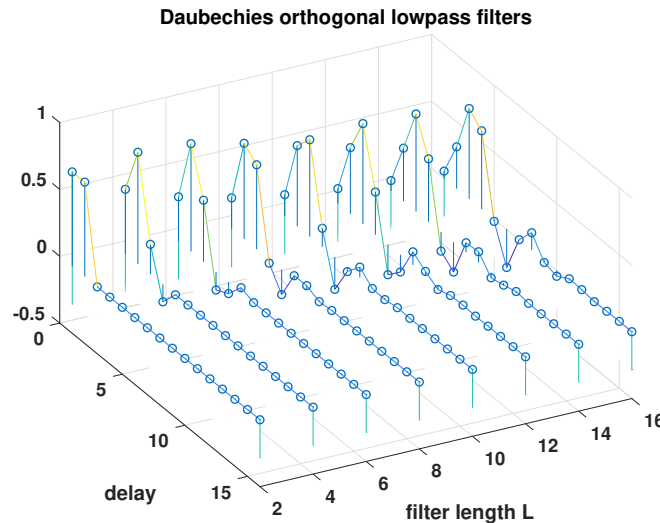
$$= \frac{\frac{17}{16} - \frac{1}{4}z - \frac{1}{4}z^{-1}}{\frac{5}{4} - \frac{1}{2}z - \frac{1}{2}z^{-1}} = \frac{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{4}z)}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{2}z)}$$



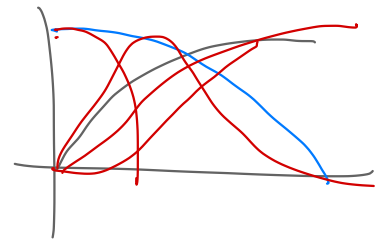
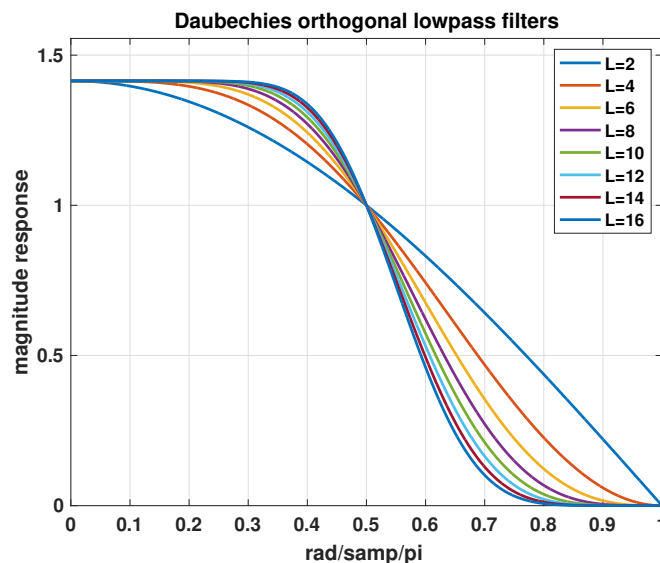
$H(z)$ stable causal $\Rightarrow H(z)$ minimum phase
 $(H(z))^n$ stable causal

Daubechies Orthogonal PR-FIR (cont.):

- The Daubechies LPF $g_1[n]$ is shown below for various L .
 - The $L = 2$ filter is simply $[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]$. Also called “Haar.”



- The magnitude responses $|G_1(e^{j\omega})|$ are shown below.
 - The response around $\omega = \pi$ gets flatter as L increases.



- The Daubechies length- L orthogonal design is abbreviated “db $\frac{L}{2}$ ” when used in the DWT. db1 is also called “Haar.”

$$W \vec{x}$$

s.t.

$$W^T W = I$$

$$W^{-1} = W^T$$

$$W^T W \neq I$$

Design

$$\tilde{W}$$

s.t.

$$\tilde{W}^T W = I$$

$\mathbb{C}^n$: vector space of column vectors.

$\underbrace{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n}_{\in \mathbb{C}^n}$ are linearly independent

$\forall \vec{x} \in \mathbb{C}^n, \exists c_1, c_2, \dots, c_n$ s.t.

$$\vec{x} = \sum_{k=1}^n c_k \vec{a}_k$$

basis for $\mathbb{C}^n$

$$\vec{x} = \underbrace{\begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \dots & \vec{a}_n \end{bmatrix}}_A \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}}_{\vec{c}}$$

$$\vec{x} = A \vec{c} \quad \text{synthesis}$$

In a special case, when $A^H A = I_n$, we have

$$\begin{bmatrix} \vec{a}_1^H \vec{x} \\ \vec{a}_2^H \vec{x} \\ \vdots \\ \vec{a}_n^H \vec{x} \end{bmatrix} = A^H \vec{x} = A^H A \vec{c} = \vec{c}$$
$$\vec{x} \mapsto \vec{c} = \underbrace{A^H \vec{x}}_{\text{analysis}}$$

In general, since A is invertible, we have

$$\vec{c} = A^{-1} \vec{x} = \begin{bmatrix} \vec{b}_1^H \\ \vec{b}_2^H \\ \vdots \\ \vec{b}_n^H \end{bmatrix} \vec{x}$$
$$(A^{-1})^H = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \dots & \vec{b}_n \end{bmatrix}$$

$\mathbb{C}^n$: vector space

$\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$: linearly indep

$\vec{b}_1, \vec{b}_2, \dots, \vec{b}_n$: linearly indep

X

$$T_{ik}^{jn} \vec{a}_k = \delta_{jle}$$

orthogonal basis

$$A = [\vec{a}_1, \dots, \vec{a}_n]$$

$$B = [\vec{b}_1, \dots, \vec{b}_n] = (A^{-1})^H$$

$$B^H A = I_n$$

$$\underbrace{\vec{x}' \mapsto \vec{c} = B^H \vec{x}'}_{\text{analysis}}$$

$$\text{st. } \underbrace{\vec{x} = A \vec{c}}_{\text{synthesis}}$$

B^H : transform
 $\vec{x}' \mapsto \vec{c}$

A : inverse transform
 $\vec{c} \mapsto \vec{x}$

Haar wavelet or db1 wavelet

ex) $B^H =$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}$$

$B^H B = I_8$

Advantage of orthogonal transform

$$\vec{x} \mapsto \vec{c} = A\vec{x}$$

$$A^H A = I_n$$

$$\begin{aligned}\|\vec{c}\|^2 &= \vec{c}^H \vec{c} = (A\vec{x})^H A\vec{x} = \vec{x}^H A^H A \vec{x} \\ &= \vec{x}^H \vec{x} = \|\vec{x}\|^2\end{aligned}$$

$$\vec{y} \mapsto \vec{d} = A\vec{y}$$

$$\vec{c} \cdot \vec{d} = (A\vec{x})^H (A\vec{y}) = \vec{x}^H A^H A \vec{y} = \vec{x} \cdot \vec{y}$$

$$A = U \Sigma V^H \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$$

$$= U \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \ddots \\ & & & \sigma_n \end{bmatrix} V^H \quad \text{s.t.} \quad \begin{aligned} U^H U &= I_n \\ V^H V &= I_n \end{aligned}$$

singular value decomposition or SVD.

$$\|A(\vec{x} - \vec{y})\| \leq \sigma_1 \|\vec{x} - \vec{y}\|$$

$$\|A(\vec{x} - \vec{y})\| \geq \sigma_n \|\vec{x} - \vec{y}\|$$

Biorthogonal PR-FIR Filters:

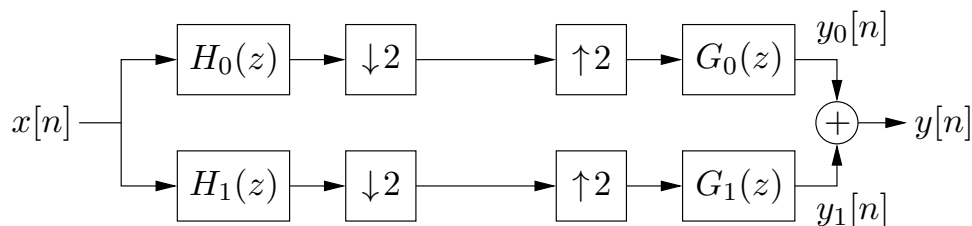
- Recall that the 2-channel PR-FIR conditions are

$$\begin{aligned} H_0(z) \text{ \& } H_1(z) &: \text{causal real-coefficient FIR} \\ \det(\mathbf{H}(z)) &= 2z^{-\ell} \text{ for } \ell \in \mathbb{Z}^+ \\ G_0(z) &= H_1(-z) \\ G_1(z) &= -H_0(-z). \end{aligned}$$

- Previously we considered “orthogonal” PR-FIR designs, with

$$\begin{aligned} \sum_n g_i[n]g_i[n-2m] &= \delta[m] \quad \& \quad \sum_n g_0[n]g_1[n-2m] = 0 \\ \sum_n h_i[n]h_i[n-2m] &= \delta[m] \quad \& \quad \sum_n h_0[n]h_1[n-2m] = 0 \end{aligned}$$

where the filterbank splits $x[n]$ into orthogonal subspaces.



- But orthogonal PR-FIR designs don't allow *linear-phase* filters (when $L > 2$). In some applications, we want linear-phase.
- For linear-phase, we can use a *biorthogonal* PR-FIR design:

$$\begin{aligned} \sum_n g_0[n]h_0[2m+L-1-n] &= \delta[m] \quad \& \quad \sum_n g_0[n]h_1[2m-1-n] = 0 \\ \sum_n g_1[n]h_1[2m+L-1-n] &= \delta[m] \quad \& \quad \sum_n g_1[n]h_0[2m-1-n] = 0 \end{aligned}$$
 under which $y_0[n]$ is generally not orthogonal to $y_1[n]$.
- Interpretation of biorthogonal: The i th analysis subspace is orthogonal to the j th synthesis subspace when $i \neq j$.

Biorthogonal PR-FIR Filters — Construction:

- The “biorthogonal” PR-FIR filterbank uses causal, real-coefficient, FIR filters $H_0(z)$ and $H_1(z)$ such that

$$H_0(z)H_1(-z) = z^{-(L-1)}F(z), \text{ with even } L \quad (3a)$$

$$F(z) = \sum_{n=-(L-1)}^{L-1} f[n]z^{-n} \quad (3b)$$

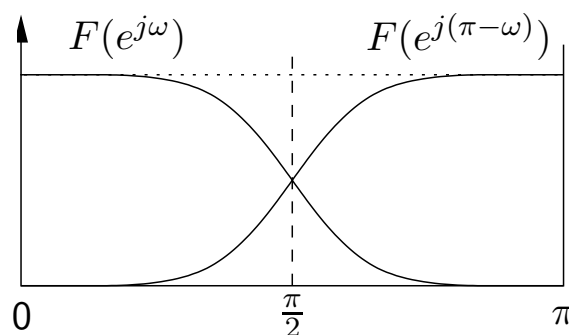
$$f[n] = f^*[-n] \in \mathbb{R}, \text{ i.e., “real, zero-phase”} \quad (3c)$$

$$f[2m] = \delta[m], \text{ i.e., “halfband.”} \quad (3d)$$

- Zero-phase halfband means $F(z) + F(-z) = 2$ (recall p. 7).
- The biorthogonal design meets the PR-FIR conditions because

$$\begin{aligned} \det(\mathbf{H}(z)) &= H_0(z)H_1(-z) - H_0(-z)H_1(z) \\ &= z^{-(L-1)}(F(z) + F(-z)) \text{ via (3a)} \\ &= 2z^{-(L-1)} \text{ via (3d)}. \end{aligned}$$

- Real-coef zero-phase halfband implies “amplitude symmetry”:
 $2 = F(e^{j\omega}) + F(e^{j(\omega-\pi)})$ via $z \rightarrow e^{j\omega}$
 $= F(e^{j\omega}) + F^*(e^{j(\pi-\omega)})$ since $(f[n] \in \mathbb{R}) \Rightarrow F(e^{j\omega}) = F^*(e^{-j\omega})$
 $= F(e^{j\omega}) + F(e^{j(\pi-\omega)})$ since $f[n] = f^*[-n] \Rightarrow (F(e^{j\omega}) \in \mathbb{R})$



Biorthogonal PR-FIR — Biorthogonality:

- Under (3), the biorthogonal filters obey (see proof next page):

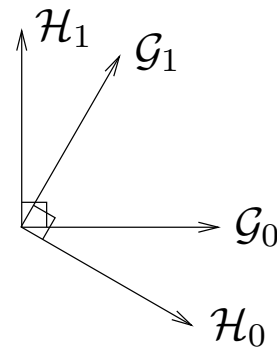
$$\sum_{n \in \mathbb{Z}} g_i[n] h_i[2m + L - 1 - n] = \delta[m] \quad \text{for } i = 0, 1. \quad (4)$$

- In addition, it can be shown that (3) implies

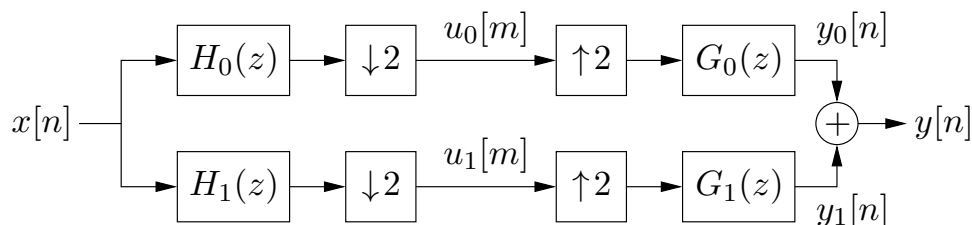
$$\sum_{n \in \mathbb{Z}} g_0[n] h_1[2m + L - 1 - n] = 0 \quad \forall m \quad (5a)$$

$$\sum_{n \in \mathbb{Z}} g_1[n] h_0[2m + L - 1 - n] = 0 \quad \forall m \quad (5b)$$

- Interpretation: (5) says that the i th synthesis space (i.e., all even shifts of $g_i[n]$) is orthogonal to the j th analysis space (i.e., all even shifts of $h_j[L-1-n]$) when $i \neq j$. And (4) ensures that the filters are properly normalized.



- Orthogonality is a special case of biorthogonality.
 - In the orthogonal case, we have $g_i[n] = h_i[L - 1 - n]$.
- So, with a biorthogonal filterbank, it is possible to make the output sequences $y_0[n]$ and $y_1[n]$ orthogonal, but in general they are *not* orthogonal.



Biorthogonal PR-FIR Filters — Proof:

- To see why the design (3) yields (4), we first write (4) as

$$c_0[2m+L-1] = \delta[m] \quad \text{with} \quad c_0[k] \triangleq \sum_{n \in \mathbb{Z}} g_0[n] h_0[k-n].$$

- In the z -domain, $c_0[k]$ simply becomes $C_0(z) = G_0(z)H_0(z)$.
- Note that $c_0[2m+L-1]$ is a 2-downsampled version of $(L+1)$ -advanced $c_0[k]$. So “ $\delta[m] = c_0[2m+L-1]$ ” becomes

$$1 = \frac{1}{2} \sum_{p=0}^1 C_0(z^{\frac{1}{2}} e^{-j\pi p}) (z^{\frac{1}{2}} e^{-j\pi p})^{L-1}.$$

Since this must hold for all $z \in \mathbb{C}$, we can also write it as

$$\begin{aligned} 2 &= \sum_{p=0}^1 C_0(z e^{-j\pi p}) (z e^{-j\pi p})^{L-1} = [C_0(z) z^{L-1} - C_0(-z) z^{L-1}] \\ &= [G_0(z) H_0(z) - G_0(-z) H_0(-z)] z^{L-1} \quad \text{recalling } L \text{ is even.} \end{aligned}$$

- Under PR-FIR condition $G_0(z) = H_1(-z)$, this becomes

$$2z^{-(L-1)} = H_0(z) H_1(-z) - H_0(-z) H_1(z).$$

- For the biorthogonal design (3), we have

$$\begin{aligned} &H_0(z) H_1(-z) - H_0(-z) H_1(z) \\ &= z^{-(L-1)} F(z) + z^{-(L-1)} F(-z) \quad \text{via (3a)} \\ &= (F(z) + F(-z)) z^{-(L-1)} = 2z^{-(L-1)} \quad \text{via (3d),} \end{aligned}$$

which matches the above manifestation of (4).

Biorthogonal PR-FIR Filters — Design:

- Recall that the biorthogonal PR-FIR filterbank uses filters

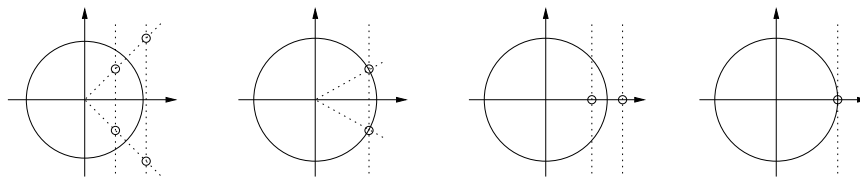
$$H_0(z)H_1(-z) = z^{-(L-1)}F(z), \text{ with even } L$$

$$F(z) = \sum_{n=-(L-1)}^{L-1} f[n]z^{-n}$$

$$f[n] = f^*[-n] \in \mathbb{R}, \text{ i.e., "real, zero-phase"}$$

$$f[2m] = \delta[m], \text{ i.e., "halfband."}$$

- To construct the filters, we build $F(z)$ (using, e.g., the Daubechies approach) and factor into $H_0(z)$ and $H_1(-z)$. Note that $H_1(-z) = G_0(z)$ due to the PR-FIR conditions.
- Because the coefficients $f[n]$ are both symmetric and real, the roots of $F(z)$ are symmetric across both the real axis and unit circle. The $2(L-1)$ roots come in groups of the form



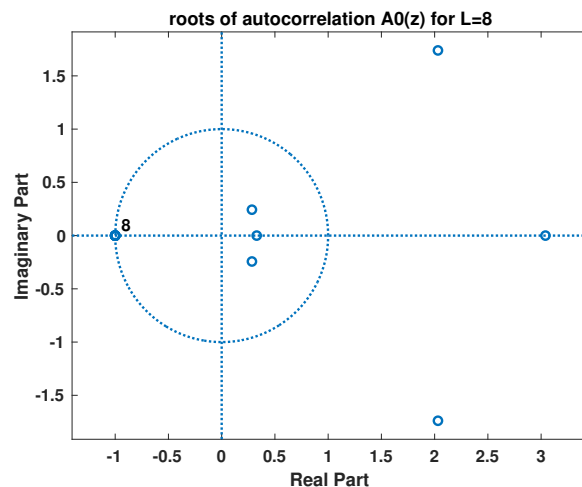
- When factoring the roots of $F(z)$ into $H_0(z)$ and $H_1(-z) = G_0(z)$, we keep the appropriate root symmetries:
 - For real-coefficient $H_i(z)$, keep complex-conjugate pairs.
 - For linear phase $H_0(z)$ and $G_0(z)$, keep roots symmetric across the unit circle, since filter coefficients are known to be symmetric about the center.

Biorthogonal PR-FIR Filters — Example:

- For biorthogonal filters, we first need a real-coef $F(z)$ such that $F(z) + F(-z) = 2$.
- For orthogonal PR-FIR filters, we created real-coef $A_0(z)$ such that $A_0(z) + A_0(-z) = 2$. It is common to use this $A_0(z)$ for $F(z)$. But note that $A_0(z)$ is more strict, in that $A(e^{j\omega}) \geq 0$.

- The Daubechies approach constructed
$$A_0(z) = (1 + z^{-1})^{\frac{L}{2}} (1 + z)^{\frac{L}{2}} \overbrace{R(z)R(z^{-1})}^{Q(z)},$$
 and found real-coef, symmetric $Q(z)$ s.t. $A_0(z) + A_0(-z) = 2$.

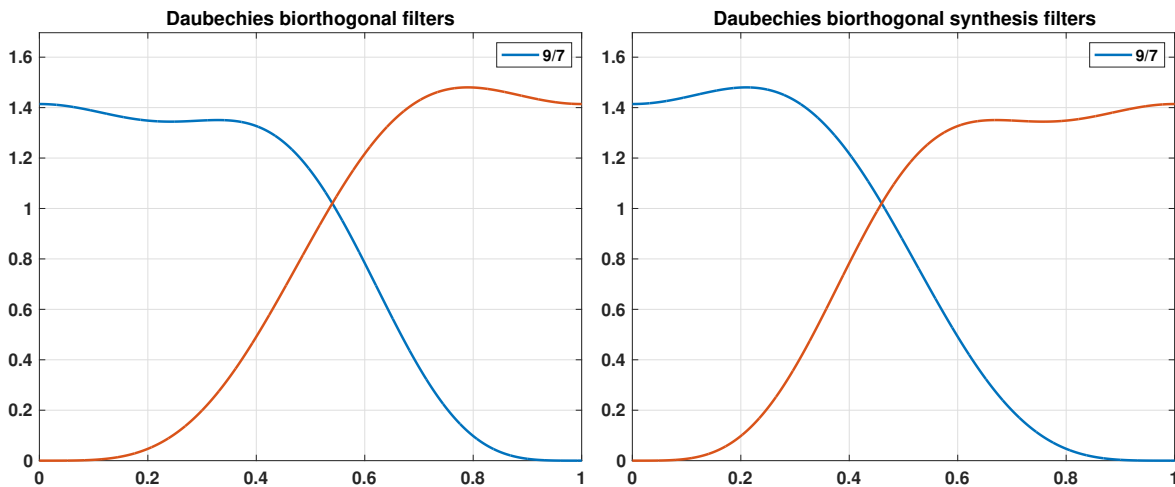
For example, when $L = 8$, this yields the $A_0(z) = F(z)$ with roots shown on the right.



- We then partition these roots into $H_1(-z) = G_0(z)$ and $H_0(z)$ in any way that obeys the desired root symmetries *and* that results in good low-pass filters.
- When normalizing the coefficients, we can use $\sum_n h_0[n] = H_0(e^{j0}) = \sqrt{2}$ and $\sum_n g_0[n] = G_0(e^{j0}) = \sqrt{2}$, or any choice such that $H_0(e^{j0})G_0(e^{j0}) = 2$.

Daubechies Biorthogonal PR-FIR Filters:

- The linear-phase “Daubechies 9/7” design creates $H_0(z)$ from 4 of the roots at $z = -1$ and the 4 complex-valued roots, and creates $H_1(-z) = G_0(z)$ from the remaining 4 roots at $z = -1$ and the remaining 2 real-valued roots.
- “9/7” signifies that $h_0[n]$ is length-9 and $h_1[n]$ is length-7.
- The filterbank magnitude responses are shown below.



- The Daubechies 9/7 design is used in JPEG 2000.
- Daubechies biorthogonal wavelets are also used in fingerprint compression standards.

Sparse Denoising

$$\vec{x} \in \mathbb{R}^n$$

$$W \in \mathbb{R}^{n \times n} \text{ discrete wavelet transform.}$$

$$\vec{x} \text{ is "smooth"} \Rightarrow W\vec{x} \text{ is "sparse"}$$



$$\vec{c} = W\vec{x} \text{ coefficient vector.}$$

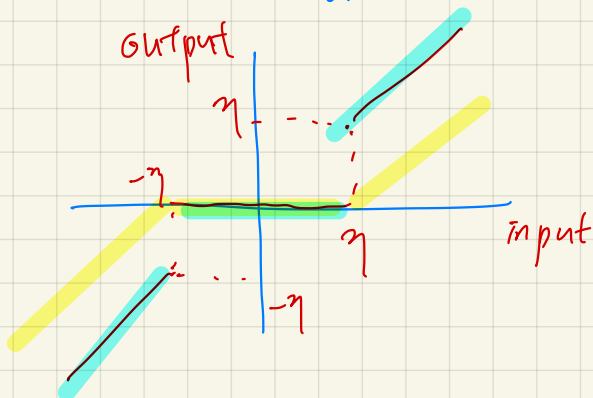
$$\|\vec{c}\|_0 = \# \text{ of } c_k \text{ s.t. } c_k \neq 0$$

$$\vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Suppose only 10% of $\vec{c}$ is nonzero.

$$[H_\eta(\vec{c})]_k = \begin{cases} c_k & \text{if } |c_k| \geq \eta \\ 0 & \text{else} \end{cases}$$

Hard-thresholding operator.



$$\vec{x} \longrightarrow \vec{c} = W\vec{x} \longrightarrow T_\eta(\vec{c}) \longrightarrow W^{-1}T_\eta(\vec{c})$$

$$\vec{x} \in \mathbb{R}^n$$

$$\min_{\vec{v} \in \mathbb{R}^n} \quad \frac{1}{2} \|\vec{x} - \vec{v}\|_2^2 + \lambda \underbrace{\|\vec{v}\|_1}_{\text{regularizer or penalty}}$$

regularizer
OR
penalty

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$\|\vec{v}\|_1 = \sum_{k=1}^n |v_k|$$

$$\|\vec{v}\|_2 = \sqrt{\sum_{k=1}^n |v_k|^2}$$

$$\|\vec{v}\|_1 \leq \sqrt{\|\vec{v}\|_0} \cdot \|\vec{v}\|_2$$

$$\|\vec{v}\|_0 = |\{k: v_k \neq 0\}| = \# \text{ nonzero entries}$$

$$\frac{1}{2} \|\vec{x} - \vec{v}\|_2^2 + \lambda \|\vec{v}\|_1$$

$$= \sum_{k=1}^n \left(\frac{1}{2} |x_k - v_k|^2 + \lambda |v_k| \right)$$

For each k ,

$$\min_{v_k} \quad \frac{1}{2} |x_k - v_k|^2 + \lambda |v_k|$$

$$\min_{v \in \mathbb{R}} \frac{1}{2} (x-v)^2 + \lambda |v|$$

Case 1 $v > 0$

$$\min_v \frac{1}{2} (x-v)^2 + \lambda v = f(v)$$

$$f'(v) = v - x + \lambda = 0$$

$$v = x - \lambda$$

Case 2 $v < 0$

$$\min_v \frac{1}{2} (x-v)^2 - \lambda v$$

$$v - x - \lambda = 0$$

$$v = x + \lambda$$

Case 3 $v = 0$

